

(3.2) Nechť V je vektorový prostor spojitých reálných funkcí s definičním oborem $[1, 5]$ a $\langle, \rangle$ je skalární součin na V daný vztahem

$$\langle f, g \rangle = \int_1^5 f(x)g(x) dx.$$

Spočítejte následující:

- ortonormální bázi podprostoru $LO\{1, x, x^2\}$ prostoru V pomocí Gramovy-Schmidtovy ortogonalizace,
- ortogonální projekce funkce $\sin(x)$ na tento podprostor, vyjádřenou jako polynom $ax^2 + bx + c$ pro vhodné koeficienty $a, b, c \in \mathbb{R}$,
- normu rozdílu $\|\sin(x) - (ax^2 + bx + c)\|$.

a) VSTUP: $\{1, x, x^2\}$

VÝSTUP: $\{u_1, u_2, u_3\}$ ON $\wedge LO\{1, x, x^2\} = LO\{u_1, u_2, u_3\}$

$\{e_1, e_2, e_3\}$

vzhledem k $\langle, \rangle$

1. normalizace: $u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{\langle 1, 1 \rangle}} = \frac{1}{\sqrt{\int_1^5 (1 \cdot 1) dx}} = \frac{1}{\sqrt{\langle x \rangle_1}} = \frac{1}{\sqrt{4}} = \underline{\underline{\frac{1}{2}}}$

2. ortogonalizace $u_2' = v_2 - \text{pr}_{\text{span}\{u_1\}}(v_2) = v_2 - \langle u_1, v_2 \rangle u_1$
 $= x - \int_1^5 \left(\frac{1}{2} x\right) dx \cdot \frac{1}{2} = x - \frac{1}{2} \cdot \left[\frac{x^2}{2}\right]_1^5 \cdot \frac{1}{2} = x - \frac{6}{2} = \underline{\underline{x-3}}$

3. normalizace $u_2 = \frac{u_2'}{\|u_2'\|} = \frac{x-3}{\sqrt{\langle x-3, x-3 \rangle}} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{x-3}{\sqrt{16}} = \underline{\underline{\frac{\sqrt{3}}{4} \cdot (x-3)}}$

4. ortogonalizace $u_3' = v_3 - \text{pr}_{\text{span}\{u_1, u_2\}}(v_3) = v_3 - \langle u_1, v_3 \rangle u_1 - \langle u_2, v_3 \rangle u_2 =$
 $= x^2 - \int_1^5 \left(\frac{1}{2} x^2\right) dx \cdot \frac{1}{2} - \int_1^5 \left(\frac{\sqrt{3}}{4} (x-3)\right) x^2 dx \cdot \frac{\sqrt{3}}{4} (x-3) =$
 $= x^2 - \frac{62}{3} \cdot \frac{1}{2} - 8\sqrt{3} \cdot \frac{\sqrt{3}}{4} (x-3) = x^2 - \frac{31}{3} - 6(x-3) = \underline{\underline{x^2 - 6x + \frac{23}{3}}}$

5. normalizace $u_3 = \frac{u_3'}{\|u_3'\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\langle x^2 - 6x + \frac{23}{3}, x^2 - 6x + \frac{23}{3} \rangle}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 \left((x-3)^2 - \frac{4}{3}\right)^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\frac{16}{3\sqrt{5}}} =$
 $= \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right) = \underline{\underline{\frac{\sqrt{5}}{16} (3x^2 - 18x + 23)}}$

Ortonormální báze podprostoru $LO\{1, x, x^2\}$ je:

$\left\{ \frac{1}{2}, \frac{\sqrt{3}}{4} (x-3), \frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \right\}$

$$\begin{aligned}
 b) \text{ pr}_{\text{span}\left\{\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{\sqrt{5}}{10}(3x^2-18x+23)\right\}}(\sin x) &= \langle u_1, \sin x \rangle u_1 + \langle u_2, \sin x \rangle u_2 + \langle u_3, \sin x \rangle u_3 \\
 &= \langle \frac{1}{2}, \sin x \rangle \frac{1}{2} + \langle \frac{\sqrt{3}}{4}(x-3), \sin x \rangle \frac{\sqrt{3}}{4}(x-3) + \langle \frac{\sqrt{5}}{10}(3x^2-18x+23), \sin x \rangle \frac{\sqrt{5}}{10}(3x^2-18x+23) \\
 &= \int_1^5 \underbrace{\left(\frac{1}{2}\right) \sin(x)}_{\cos 1 - \cos 5} dx \cdot \frac{1}{2} + \int_1^5 \underbrace{\left(\frac{\sqrt{3}}{4}(x-3) \cdot \sin(x)\right)}_{(-1,493) \cdot \frac{\sqrt{3}}{4}(x-3)} dx \cdot \frac{\sqrt{3}}{4}(x-3) + \int_1^5 \underbrace{\left(\frac{\sqrt{5}}{10}(3x^2-18x+23) \cdot \sin(x)\right)}_{(-0,125) \cdot \frac{\sqrt{5}}{10}(3x^2-18x+23)} dx \cdot \frac{\sqrt{5}}{10}(3x^2-18x+23) \\
 &= \underbrace{\frac{1}{2}(\cos 1 - \cos 5)}_{0,064} \cdot \frac{1}{2} + \underbrace{(-1,493) \cdot \frac{\sqrt{3}}{4}(x-3)}_{-0,646x + 1,939} + \underbrace{(-0,125) \cdot \frac{\sqrt{5}}{10}(3x^2-18x+23)}_{-0,053x^2 + 0,315x - 0,406} =
 \end{aligned}$$

$$\boxed{\text{pr} \approx \underbrace{-0,053x^2}_a - \underbrace{0,331x}_b + \underbrace{1,597}_c}$$

$$\begin{aligned}
 c) \| \sin(x) - (-0,053x^2 - 0,331x + 1,597) \| &= \sqrt{\int_1^5 (\sin(x) - (-0,053x^2 - 0,331x + 1,597))^2 dx} = \sqrt{0,1019} \\
 &\approx \underline{\underline{0,319}}
 \end{aligned}$$